JAYA SHARMA

Senior Software Engineer
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SUMMARY

Backend Software Engineer with 7+ years of experience architecting and owning scalable distributed systems and high-throughput backend services end-to-end. Proven track record leading real-time fintech platform delivery using Java, Spring Boot, and Kafka, from design through production. Strong foundation in system design, clean architecture, DSA, and cross-team technical ownership. Currently building expertise in AI application development, including LLM integration, RAG pipelines, embeddings, vector databases, and Spring AI.

SKILLS

Languages: Java (Primary), Python, Dart
Core Concepts: Data Structures & Algorithms, OOP, DBMS, SOLID Principles
Frameworks: Spring Boot, Play Framework, Flutter, Android
Data & Messaging: Kafka, RabbitMQ, MongoDB, Firestore, MySQL, Cassandra, SQLite, SQFLite
Platforms & Observability: Docker, Splunk, Grafana, Prometheus, UrbanCode, CyberArk
AI / GenAI: LLM APIs, Prompt Engineering, RAG, Embeddings, Vector Databases, Semantic Search
Tooling: VSCode, IntelliJ, STS, Eclipse, GitHub, JIRA, Confluence, Git

EXPERIENCE

Lloyds Technology Centre — Senior Software Engineer *Dec 2023 – Present*
Rates Re-architecture, Front-Office Workflow

- Co-led the architecture and redesign of a real-time risk, PnL, and pricing platform for fixed-income assets (Swaps, Gilts); owned production rollout in partnership with trade strategists, cutting pricing turnaround by around 50% and speeding trade decisions for the desk.
- Architected and built a low-latency data pipeline on Kafka and REST APIs, scaling throughput to handle high-volume pricing data while holding sub-10ms p99 latency on latency-critical paths, enabling real-time pricing during peak trading volume.
- Owned observability strategy for the platform, implementing a Splunk/Grafana stack covering 10+ production services; cut incident detection/triage time by 70–80% and reduced recurring production issues by 90%, saving 80+ engineering hours per quarter.
- Drove infrastructure and CI/CD improvements in partnership with DevOps/SRE/Network teams, provisioning and tuning Linux servers for deployments and mentoring 3 junior engineers on coding/release best practices.

DBS Bank — Software Engineer *Jan 2022 – Nov 2023*
CRANE Project — Credit Risk Analysis

- Led enhancement of a credit-risk model (R) in partnership with strategists, scoring loan-repayment risk using financial, network, and buyer-supplier data; improved model accuracy, directly informing lending decisions across a portfolio of thousands of counterparties.
- Owned backend and frontend delivery for risk-platform enhancements, partnering directly with risk analysts to translate requirements into production-ready features.

IMUMZ — Software Engineer *May 2021 – Dec 2021*
Mindful Pregnancy and Parenting App

- Owned end-to-end delivery of referral, health-section, and subscription-program features on an Onion Architecture, from design through release.
- Designed the Firestore data model and query strategy, applying pre-caching and minimal-build-cost patterns to keep cold-start times under 2s and improve app responsiveness.
- Implemented Firebase Cloud Functions for backend processing of subscription and health-data workflows.

Vassarlabs IT Solutions — Software Engineer *Jan 2019 – Apr 2021*
Uniapp Framework

- Architected and built, from scratch, a generic Flutter mobile framework adopted as the foundation for 10+ production apps company-wide, cutting new-app bootstrap time by 90% and establishing reusable engineering standards for the mobile team.
- Authored and championed reusable product-level modules adopted by multiple internal teams, reducing duplicate engineering effort across projects.
- Built custom offline maps and navigation, enabling core functionality with zero connectivity for users in regions of Andhra Pradesh.

PROJECTS

RAG-based Financial Knowledge Assistant — Java, Spring Boot, Spring AI, LLM, Vector Database

- Built a hands-on RAG pipeline to ingest documents, generate embeddings, store vector representations, retrieve relevant context, and generate LLM-powered responses.
- Explored semantic retrieval and metadata-based filtering to improve retrieval relevance and control the context supplied to the LLM.
- Implemented the AI workflow using a Java/Spring Boot backend and Spring AI, focusing on practical integration of LLM and retrieval components.

EDUCATION

B.Tech, Computer Science & Engineering — IIIT Vadodara *2015 – 2019*
CGPA: 8.1

RESEARCH & PUBLICATIONS

- "Exponentially Weighted Random Forest" — 8th International Conference on Pattern Recognition and Machine Intelligence (PReMI), 2019. View on SpringerLink. Improved efficiency across multiple datasets, with results comparable to or exceeding prior RF variants on classification and regression tasks.